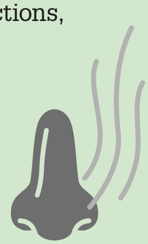
Sensing the World

To survive in our environment, we must be able to react to, and interact with, stimuli produced by physical, chemical, and biological phenomena—sights, sounds, smells, tastes, and touches. Sensors in the body pick up these signals and send them to the brain for deciphering.

Senses

Each sense has its own set of detectors. Most are localized in a specific area of the body, except for touch, which is spread all over the skin, as well as inside the body. Although the neurons and receptors for each sense are largely dedicated to that sense alone, they can sometimes overlap. Sensory information continuously bombards the brain, but only a fraction of the input reaches consciousness. Even so, the "unnoticed" information can still guide our actions, particularly in the case of our sixth sense, proprioception, which relays information about the body's position in space.



YOUR **SENSE** OF **SMELL IMPROVES** WHEN YOU ARE **HUNGRY**

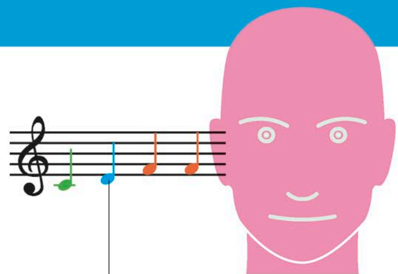
Touch
Thought to be the first sense to develop in the womb, touch neurons respond to pressure, temperature, vibration, pain, and light touch. Touch is how humans make physical contact with the environment and with each other.

Hearing
Sound waves in the air are collected by the ear and transmitted into the skull, where they are turned into electrical impulses by the cochlea. Hearing is the most developed of the senses at birth but is only complete by the end of the first year.

Sight
Sight involves sensors at the back of the eye that turn light into electrical signals. These are transported to the back of the brain, where they are converted into colors, fine details, and motion. We perceive objects in as little as half a second.

SYNESTHESIA

Synesthesia is a condition where a stimulus may be interpreted by two or more senses at the same time. In its most common form, a person sees a number or word as a color. Each synesthete will have its own color associations. Almost any combination of senses can be affected. Combinations of three or more senses are rare.



Each note is associated with a different color

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